

Remote Access Architecture: Non-LAN Connectivity

How RustDesk Enables Remote Access Across WAN / Internet Firewalls

Protocol: RustDesk P2P / Relay Protocol | **Security Model:** E2E Encryption (NaCl / Ed25519) | **NAT Traversal:** STUN & Rendezvous (hbbs / hbbr)

CORE ARCHITECTURAL CONCEPT:

A remote user not connected to the local network (LAN) accesses this machine via **Outbound Rendezvous & Relay Signaling**. Because this system initiates an *outbound* connection to a public or self-hosted rendezvous server (hbbs), external clients can discover, punch through local NAT/firewalls, or relay traffic to this host without requiring port forwarding, static public IPs, or local network visibility.

1. End-to-End WAN Connection Lifecycle

When an external remote client connects from outside your LAN (e.g. from home, cellular data, or another office), the connection executes through four sequential phases:

Phase	Mechanism	Technical Execution Details
1. Outbound Registration	Heartbeat / Signaling	When this machine boots, the RustDesk Windows Service (PID 4872) establishes an outbound TCP/UDP socket to the Rendezvous Server (hbbs). Standard routers allow outbound traffic automatically without port forwarding.
2. ID Lookup & Pairing	Rendezvous Broker (hbbs)	The external user inputs this host's unique 9-digit or customized RustDesk ID into their client. The client queries hbbs, which maps the ID to this host's active public IP and ephemeral UDP port.
3. NAT Traversal (STUN / Hole Punching)	Direct P2P UDP Tunnel	hbbs instructs both the external client and this host to send simultaneous UDP packets to each other's public endpoints. This punches holes through both NAT firewalls, establishing a direct, low-latency peer-to-peer session .
4. Fallback Relay Routing (hbbr)	Encrypted Relay Server	If symmetric NAT or strict enterprise firewalls block direct UDP hole punching, both peers connect outward to the Relay Server (hbbr). The relay blindly forwards the end-to-end encrypted stream between the endpoints.

2. Pre-Login Screen Interaction & Authentication Flow

After the WAN tunnel is established, the remote user must pass two security gates to access the operating system:

- Gate 1: RustDesk Access Authorization:** The remote user provides the permanent password, dynamic one-time password (OTP), or asymmetric public key previously configured on this host. Once verified, RustDesk grants video and input stream access.
- Gate 2: Windows Secure Desktop / Winlogon:** Because RustDesk runs as LocalSystem, the remote client is presented with the native Windows Login / Lock screen. The remote user issues a simulated Ctrl+Alt+Del (SAS keystroke), inputs their Windows username and password/PIN, and initiates a Windows interactive session (spawning Explorer and user apps).

3. Comparison: Non-LAN WAN Access Methods

Access Method	Prerequisites	How Inbound Traffic Reaches Host	Firewall Config Needed
RustDesk ID (Default)	Host RustDesk ID & Password	NAT Hole-Punching via hbbs or Relayed via hbbr	None (Outbound only)
Direct IP Connection	Public Static IP + Port	Direct TCP/UDP incoming socket to host	Router Port Forwarding (Ports 21115-21119)
VPN / Mesh Overlay (Tailscale/WireGuard)	VPN Client / Node Joined	Encrypted virtual overlay subnet (treats WAN as local)	None (Outbound VPN tunnel)

Takeaway: A remote user does not need to be on your local network, nor do they need your local IP address. They only need the target machine's **RustDesk ID** and **access password**. The RustDesk infrastructure brokers the connection across the internet seamlessly through outbound NAT traversal.